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1 Introduction

The ground segment is a key component to the success of any CubeSat mission, whether it is based on a constellation, a swarm, or a single spacecraft. The ground segment gives support to the space segment, relaying payload data to the user segment and managing Tracking, Telemetry, and Command (TT&C) for both the platform and the payload. The increasing demand for high bandwidth and the constant improvement of modulation and codification schemes set the goal of maximum flexibility for the ground segment. The traditional approach of distributed control centers is not generally possible within the budget constraints of a CubeSat mission, and all operations will be usually performed from the same facility. This is another limitation of the situation when there is a need to provide support to several missions and modes using the same ground station (GS).

In this chapter the typical architecture of a modern ground segment will be introduced, using cost-effective, cutting-edge technology and emphasizing flexibility, presenting state-of-the-art solutions that can improve the operations workflow.

2 Ground segment overview

The ground segment comprises all the Earth-based elements that give support to one or several spacecraft during the course of a mission (or several, even simultaneously) [1, 2]. The main elements are the ground stations, which provide the communication interface, operation centers from where spacecraft are managed, and ground networks that provide interconnection between the elements. An overview of the typical architecture can be found in Fig. 1.

2.1 Functionality

The ground segment enables the management and operation of the involved space segments. It also has to provide access to the required information and science data from the user segment side. In summary, it has to provide the following functionalities to the interested parties:

- TT&C
 - Communication and control
 - Housekeeping and telemetry acquisition and processing to determine the status of the spacecraft

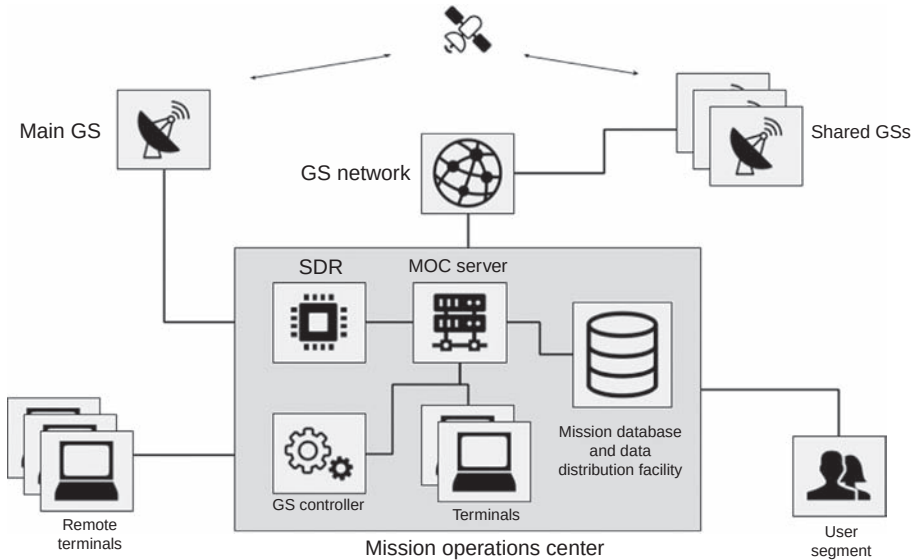


Fig. 1 Ground segment architecture.

- Forwarding of requests from the user segment
- Scientific/payload data gathering
- Ground station control
 - Configuration for transceivers and radios
 - Doppler corrections in real time
 - Pointing, tracking, and ranging (if required) of spacecraft
 - Management of ground stations network, in case more than one GS is available
- Mission planning
 - Generation of operation plans
 - Operators access to handle conflicts
 - Execution of schedules according to the approved plans
- Operators interface
 - Visualization of the spacecraft status and acquired data
 - Remote access to the ground segment for off-site operations
 - CCTV of the outside facilities for surveillance
- User segment access
 - Access to the scientific/payload data from stakeholders' terminals
 - Requests for scientific/payload operations

2.2 Architecture

The traditional approach of distributed facilities like ground station facilities (GSF), flight dynamics facilities (FDF), and mission operations facilities (MOF) is not the usual architecture for the stringent budget-constrained CubeSat missions. More typically a single facility with even a single server is the main topology used to manage all the required hardware, interfaces, and services [3]. One or more ground stations will

be connected to the main facility that is called the Mission Operations Centre (MOC). All the operations are generally managed from this location. The disadvantages of using a single ground station include limits in the ability of simultaneously communicating with more than one spacecraft and restrictions to CubeSats below GEO orbits as they are typically unable to mount far-ranging parabolic dishes owing to their expense and complexity [4].

A network of ground stations can be used to operate the spacecraft or the constellation, using in-house or external services. This is a rising option that provides extended access and can save an important amount of the ground segment costs, monetizing the idle time of the ground stations. However, this generally does not provide the same security, reliability, and latency as classical ground station networks [5].

3 Ground station

Space radiocommunications are possible, thanks to the ground stations. The elements of this crucial portion of the ground segment must be wisely selected to attain reliable performance during the operation of a CubeSat. Even a small deviation from the requirements could lead to an unfavorable result. A brief summary of the most important elements (represented in Fig. 2) of a ground station will be introduced to help the reader to make the right choices during the design phase of the ground segment for a CubeSat mission.

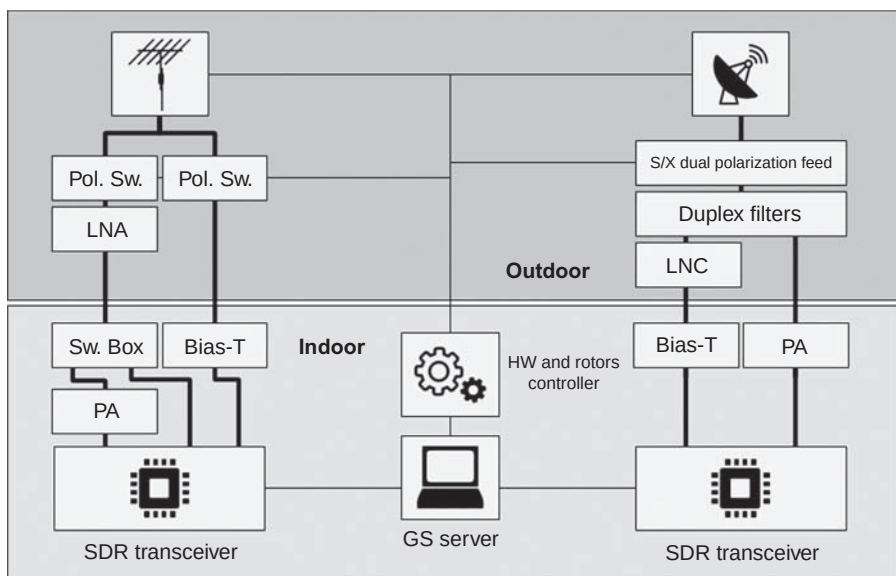


Fig. 2 Ground station block diagram.

3.1 RF hardware

3.1.1 Antennas

The main bands for space radiocommunications with CubeSats are VHF, UHF, L band, and S band. Less common but in growing trend are X band and Ka bands [3]. A widely adopted choice for **VHF and UHF** bands is the cross Yagi-Uda antenna, which usually provides great results. Making use of phasing networks, multiple polarizations are available with a single antenna; even remote configuration is possible if a polarization switch is available. The performance depends on the number of elements, at the cost of bigger and heavier antennas. To provide maximum performance on the reception, two antennas can be used for each band, providing **polarization diversity** (simultaneous reception on multiple polarizations), at the expense of having several reception chains, and avoiding the need of an attitude control system aboard the spacecraft that guarantees polarization match with the ground station antennas during contacts. Circular polarization is a common choice for CubeSats, since it guarantees communication regardless of the orientation of the spacecraft with respect to the ground station [3]; if the spacecraft is tumbling, either right-hand circular polarization (RHCP), left-hand circular polarization (LHCP), or linear polarization will be received, with a maximum polarization loss factor of 0.5 (−3 dB).

For L band and higher frequencies, helical antennas or parabolic reflectors are the main options. The decision depends on the requirements of the mission and the link budget; high gain and high data rates generally require a parabolic reflector dish. A parabolic reflector will need a feed system to capture the radio frequency wave and guide it to the transceiver through a coaxial cable or a waveguide or vice versa for the transmission. It is advisable that this dish feed supports configurable polarizations without the need of a setup modification. A complex mission may require support for multiple bands, for example, S band and X band. In this scenario a multiband dish feed is the best choice since it avoids the expensive solution of having multiple antennas. Mounting multiple dish feeds on a single parabolic reflector increases the complexity of the system significantly and does not allow optimization at any frequency, since multiple feeds negatively affect the radiation pattern of the system.

3.1.2 Filtering

The use of RF filters in some scenarios is essential. Adjacent commercial services, such as broadband cellular networks, can interfere with the communications. The tight margins of CubeSat communications typically leave little room for extra losses. In the case of full duplex communication using a single band, a duplexer based on ceramic or cavity filters can provide the required channel multiplexing while having a low impact on the overall communication system. If rigorous filtering is required, a carefully designed cavity waveguide filter with walls plated with a thin layer of gold, to improve surface conductivity, provides the lowest losses, high selectivity, and linear group delay [6].

3.1.3 Low-noise amplifiers and down-converters

Among many other limitations, power constraints set the pace of CubeSat mission operations. Very low power is available for transmissions; therefore a very weak signal will be captured by the antennas at the ground station, and this will have to overcome the feed line losses experienced in the system before the receiver. To overcome this situation a low-noise amplifier, that is, a variety of electronic amplifiers with low impact in the signal-to-noise ratio, must be placed close to the antennas or reflector feed to keep the overall noise figure as low as possible. The lower the noise figure of the amplifier, the better the signal reception. Numerous commercially available, low-noise amplifiers (LNAs) are suitable for amateur and nonamateur bands routinely used by CubeSats. They often include a voice-operated exchange (VOX) switch, enabling the LNA to switch between transmission and reception detecting the carrier on the feed line and avoiding the need of a push-to-talk (PTT) line from the receiver. Frequently, this switch consists in a RF relay with a higher-than-desired operating delay (plus a bounce period) and a limited life span. Hence a solid-state RF relay activated by a PTT line is preferable, as it will fairly improve the switching time (useful in half-duplex scenarios) and provides practically unlimited life span.

In the case of S band and higher frequencies, the use of a low-noise downconverter is convenient, since it will not only amplify the signal but also convert it to a lower frequency without a severe impact on the signal-to-noise ratio and avoiding high losses during its travel to the receiver through the transmission line, usually a coaxial waveguide. Care must be taken when adjusting the gain of the low-noise amplifier or downconverter to avoid saturation on the reception, particularly with short distances between the antenna and the transceiver. Another important consideration is the impedance of the transmission lines. Some downconverters are designed to work in bands used to feed satellite TV receivers and therefore require transmission lines with a characteristic impedance of 75Ω .

3.2 Transceivers, modems, and software-defined radios

The required architecture to support TT&C in CubeSat missions was commonly built around the robust but outdated combination of a terminal node controller (TNC) (a hardware component designed for packet radio) and a commercial transceiver. This approach has some drawbacks, since the GS can support only the bands allowed by the transceiver (usually VHF, UHF, and L band) and the modulation/codification schemes (MODCODs) and framing modes supported by the TNC, making hard to impossible to support different spacecraft and missions [3]. To overcome these limitations, software-defined radios (SDR) introduce a paradigm shift where transmission and reception chains are fully implemented by software (e.g., filtering, modulation, and mixing). This scheme allows developers to reconfigure the communication system to give support to a wide variety of changing protocols and, hence, multiple spacecraft. Now the only required hardware is a RF front end compatible with the working bands [7, 8]. Typical figures of merit of a modern CubeSat ground station built for amateur bands are included as a reference in [Table 1](#).

Table 1 Figures of merit of a typical CubeSat ground station.

	Frequency bands (MHz)	Antenna gain (dBi)	Preamplifier gain and NF (dB)	EIRP/G/T	Polarization
UHF uplink	435–438	16.2	–	32.51 dBW	RHCP/LHCP
UHF downlink	435–438	16.2	25 0.7	–3.73 dB/K	RHCP/LHCP
S-band uplink	2400–2450	31.35	–	40.13 dBW	RHCP/LHCP
S-band downlink	2400–2450	31.35	28 0.4	12.90 dB/K	

During the contacts with the spacecraft, Doppler effects must be compensated. When working with SDRs, this operation is easier than ever, since no communication with external hardware is required: just another stage in the transmission/reception chains, as long as the required bandwidth is captured. Monitoring of the spectrum is also a simple task now, without the need of external hardware like a spectrum analyzer connected to the reception chain with a RF splitter that will also cause harm to the received SNR. A single GS can be now part of the ground segment of multiple missions or a ground stations network, that is, a much more cost-effective solution than building several ground stations for a multimission project. The advent of software-defined radios has not fully replaced hardware-based solutions yet, even when it has a smaller footprint than hardware-defined radios, since it tends to be more power demanding. The use of SDRs on the space segment is more challenging, and ground stations for CubeSat missions usually match radios from the space segment.

3.3 Mechanical elements and pointing

Ground stations must track the spacecraft during the contacts, precisely pointing the antennas due to the motion of the spacecraft along the orbit. For this purpose a set of rotors mounted over a reliable structure are required. Yagi-Uda antennas are usually mounted over a metal cross boom that in turn is attached to the rotors. This metal cross boom could interfere with the radiation pattern of the Yagi-Uda antennas. As such, a glass fiber-reinforced polymer (GFRP) or carbon fiber-reinforced polymer (CFRP) cross boom is a better option since it will not cause any harm to the radiation patterns. This setup also gives the possibility of mounting a parabolic reflector over the cross boom, increasing the capabilities of a single ground station.

Several commercial rotors are available on the market [3], but this choice should be well pondered. Tracking speed and pointing precision depend on the mission requirements and the orbit selected. Higher-frequency bands (S band, X band, and upper) may require a better pointing accuracy, since a higher gain will be needed to compensate the increase in free-space path loss, and therefore bigger antennas with narrower

beam widths will be used. Meanwhile, very low orbits, mainly under the ISS, will require a very high angular speed for the rotors to keep track of the spacecraft. Precaution must be taken also in harsh environments where adverse weather conditions (e.g., wind and ice) could cause damage or malfunction of the GS. An overzealous estimation of the wind conditions can jeopardize the facilities if the brake torque of the rotors is exceeded or the GS is operated over the nominal wind pressure stipulated by the rotors manufacturer; this requires some calculations for each setup since it depends on the antennas wind surface. Ice loading on the antenna surfaces is definitely also an impediment for the operation of a ground station, particularly in the case of parabolic reflectors. In such locations, extra protection can be provided by a radome, an enclosure that protects the structure and the antennas while being effectively transparent for electromagnetic waves in the radio spectrum.

3.4 Considerations for high frequency bands

X band through Ka-band radiocommunication is gaining more traction in recent CubeSat missions as higher data rates are an indispensable requirement in contemporary complex missions. These bands offer more bandwidth and are arguably quite less congested, but these advantages come with some drawbacks that present a challenge for low-budget CubeSat missions. Atmospheric absorption increases when moving to shorter wavelengths requiring more power, which is a precious resource on board a CubeSat. Other nuances to consider are that higher frequencies antennas offer similar gain with a smaller aperture, making the pointing budget an issue, and RF and baseband hardware required to transmit and receive in these bands are usually too costly for typically budget-constrained CubeSat missions.

4 Ground segment software

It is not an overstatement to say that a massive endeavor in software development is required in any contemporary space project. A thorough study and a wise design of the software architecture are crucial for the success of the mission. Software is the glue that keeps together all the pieces of a modern ground segment for CubeSats, usually backed by open-source frameworks, underpinning the success of the mission providing high automation possibilities and agile development of new features. These aspects ring true for flight software, ground data systems, and software systems. The architecture of a typical ground segment software system is shown in [Fig. 3](#).

4.1 Mission control software

The core of the operations is the Mission Control Software (MCS). This vital piece of the ground segment will provide the required tools for telemetry download, telecommand upload, monitoring of the status of the spacecraft, data distribution and visualization, and operation planning. The MCS serves as the main interface for the

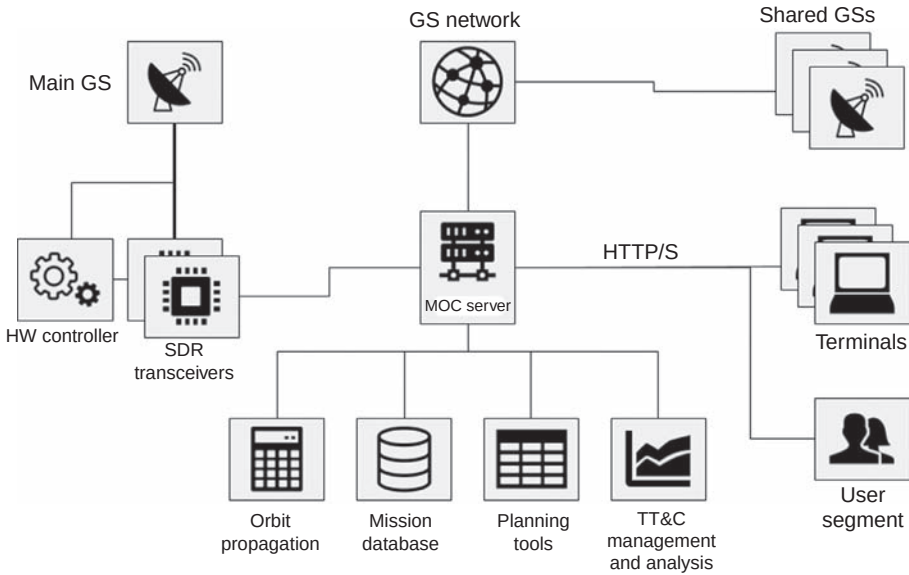


Fig. 3 Ground segment software architecture.

operators to control the spacecraft and will have to present the information in a condensed and human-friendly way, to allow the operators to take the right decisions quickly and with no place for confusions [1, 2]. A distributed, nonmonolithic, and service-oriented architecture is encouraged. This will allow the service to scale according to the demands, either more spacecraft (constellations and swarms), more operators, or more users consuming data. Another advantage of this scheme is the possibility of remote access to the MCS for off-site operations, giving the maximum flexibility to the operators.

4.2 Orbit propagation

An accurate computation of the orbital path of a spacecraft is a requirement of any space mission. Operators need to know the position of the spacecraft days or weeks in advance to plan a schedule of contacts. The ground stations have to allocate the available slots for the selected contacts in accordance with this schedule. During a pass the rotors must perform a precise tracking of the spacecraft to harness the full potential of the antennas and communication systems. Next the Doppler effect, which changes dynamically, must be compensated during a contact, so the transceiver or SDR must ingest this information in real time. The traditional role of the Flight Dynamics Facility has to automate these processes through the use of software systems that provide reliable and updated information to the rest of the ground segment. Several free tools are available for this purpose [9–11].

4.3 Ground station control

During the contacts, all the hardware must work as a perfectly oiled machine. The information provided by the orbit propagation software will be used to interact with the rotors controller to perform a smooth track of the spacecraft [12]. As previously mentioned the Doppler effect must be compensated on the reception chain in real time, and this calculation will be computed also using this information. Before the contact, all systems must be checked to generate an alarm with enough time to take actions. The antennas will be waiting for the spacecraft at the acquisition of signal (AOS) azimuth point. During the tracking phase the strategy of rotors displacement should take into account the limitations of the angular speed. Satellites in LEO orbits and lower, as is the case of many CubeSat missions operating at altitudes lower than the International Space Station, can be difficult to track. After the end of the contact referred to as loss of signal (LOS), the software must take care of parking the ground station in a safe position to prepare for the next track.

5 Ground segment operation

There is little room for improvisation throughout a space mission. All contacts with the spacecraft and every single telecommand sent must be preceded by strict planning and revision. The MCS can be an invaluable tool to simplify the operations and make planning and execution more effective and lower-risk tasks. Software tools often tailored for each mission are a key element of daily operations, supported by strict policies and constant reviews.

5.1 Operation planning

The passage of a CubeSat over a ground station is a precious resource, particularly in LEO orbits where contact windows last for about 10 minutes, a few times per day. The MCS must provide the right tools to simplify the routine operation of mission planning based on orbit propagation computations and contacts prediction [1, 2].

The generation of a schedule by the MCS should ideally be an unattended task, where the operators introduce the inputs (e.g., involved ground stations and spacecraft, time constraints, priorities, and payloads) and the software returns a simplified view (e.g., calendar and Gantt chart) of what is going on during the time frame under revision, how the slots were allocated, and can identify possible conflicts at a glance. The operator's task is to review the schedule and decide how to wisely manage the possible conflicts or hazardous situations. For example, during a multispacecraft mission, two contacts of different spacecraft can overlap at the same time slot, and one of them has to be discarded or reallocated on any other free slot. Information generated by the MCS during the planning phase can be used not only for ground segment management but also for telecommands and schedule generation, ready to upload to the spacecraft.

5.2 Contact execution

Following policies and procedures is essential to obtain professional results. Consistency in practices will facilitate effective execution of the space mission. At least two operators are suggested as a best practice for each contact. One should take care of the ground station elements such as rotors, transceivers, controllers, and software status. As a general rule, a low-level overview of the involved hardware is useful to rapidly debug possible issues. Also a plot showing the power spectral density estimation (RTSA-like view) is a very helpful tool to give an insight of what is happening at the physical layer. The other operator will drive the MCS to command the spacecraft. Before each contact opportunity, operators should have a roadmap of the actions to perform (a contact sheet, e.g., with step-by-step instructions) as a sequence of events (SOE). All the decision-making and streamlining must be embodied in the procedures. These guidelines must reflect all possible paths that operators must follow, for example, in case of a contingency; for example, a critical level of a battery could be a no-go decision, making it better to wait until the nominal level is restored. After each successful contact, operators must record in the logbook all the performed actions, the obtained results, and the issues found if an anomaly occurred. This applies to both spacecraft and GS. Future operations, particularly related to anomaly research, will find this information very useful to gain introspection and pinpoint to possible procedural flaws.

5.3 Automation

The main aspiration of any ground segment designer should be to make the operators' life as easy as possible. All procedures are susceptible to be automated, with the exception of revision. A human operator should perform tasks associated with exceptions and revisions. As mentioned earlier, ground station planning and spacecraft schedule generation automation provide numerous advantages, since this repetitive task is prone to human errors. Using automation, the ground segment and space segment operations are fully interconnected and can be managed and reviewed using the same tools. If the ground station is fully automated, the operators' primary task is monitoring its status and possible warnings. Low-risk routine operations like periodic housekeeping downloads outside canonical working hours are ideal for automation, allowing the Mission Control Centre to not have to be staffed around the clock. Ground station status can be monitored in the background, sending the appropriate alarms to the operators to be aware of possible issues beforehand and allowing to make a contingency plan for the operations. The level of targeted automation is related to the mission complexity, but most CubeSat missions can greatly benefit from some level of automation.

References

- [1] Y.J. Kenneth, *Satellite Communications. Network Design and Analysis*, Artech House, 2011.
- [2] J.R. Wertz, D.F. Everett, J.J. Puschell, *Space Mission Engineering: The New SMAD*, first ed., Microcosm Press, 2011.

- [3] Mission Design Division, Ames Research Center, NASA, Small Spacecraft Technology State of the Art, NASA Technical Memorandum, 2018.
- [4] M. Schmidt, Ground Station Networks for Efficient Operation of Distributed Small Satellite Systems, *Ausgezeichnete Informatikdissertationen*, 2011.
- [5] T. de Cola, H. Ernst, M. Marchese, Performance analysis of CCSDS file delivery protocol and erasure coding techniques in deep space environments, *Comput. Netw.* 51 (2007).
- [6] L.A. Belov, S.M. Smolskiy, V.N. Kochemasov, *Handbook of RF, Microwave, and Millimeter-Wave Components*, first ed., Artech House, 2012.
- [7] T.F. Collins, R. Getz, D. Pu, A.M. Wyglinski, *Software-Defined Radio for Engineers*, Artech House, 2018.
- [8] GNURadio, <https://www.gnuradio.org>. (Accessed December 2019).
- [9] Predict, <https://www.qsl.net/kd2bd/predict.html>. (Accessed December 2019).
- [10] GPredict, <https://github.com/csete/gpredict>. (Accessed December 2019).
- [11] Space-Track, <https://www.space-track.org/auth/login>. (Accessed December 2019).
- [12] Hamlib, <https://hamlib.github.io/>. (Accessed December 2019).